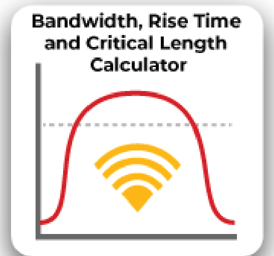
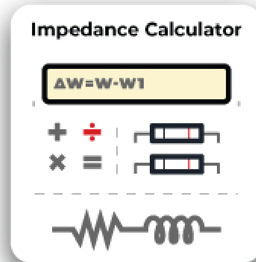
SIERRA CIRCUITS

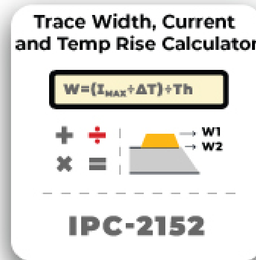
FLEX

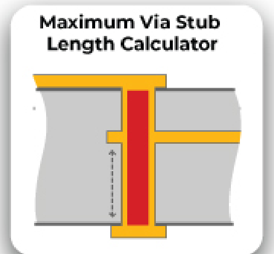
DESIGN GUIDE

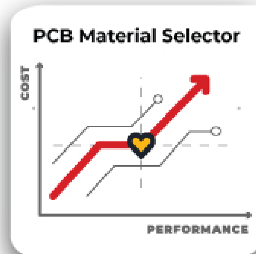
The Designer's Toolkit

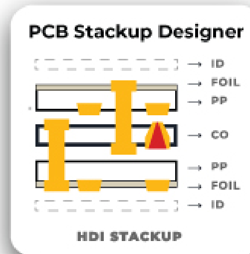


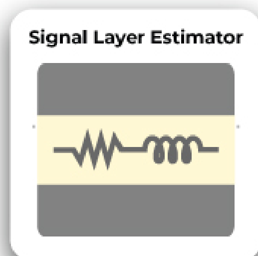


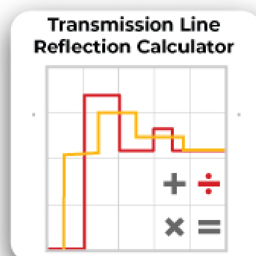


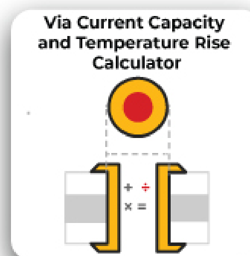


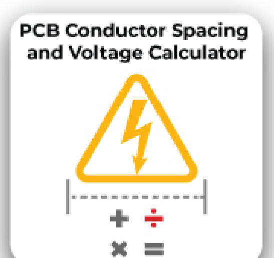


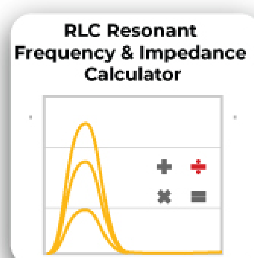


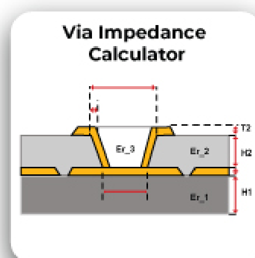


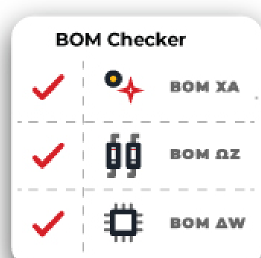












TRY TOOLS

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Chapter 1: Flex PCB and its materials

1.1 What is a flex PCB?

Flex PCBs with very thin substrates and a high level of bendability offer superior tensile strength. These circuits are popular in any high-end electronic device, such as medical devices, fitness wearables, cameras, and smartphones.

1.2 Types of flex PCBs

Flexible PCBs are classified into single-sided, double-sided, and multi-layered boards.

1.2.1 Single-sided flexible circuit boards

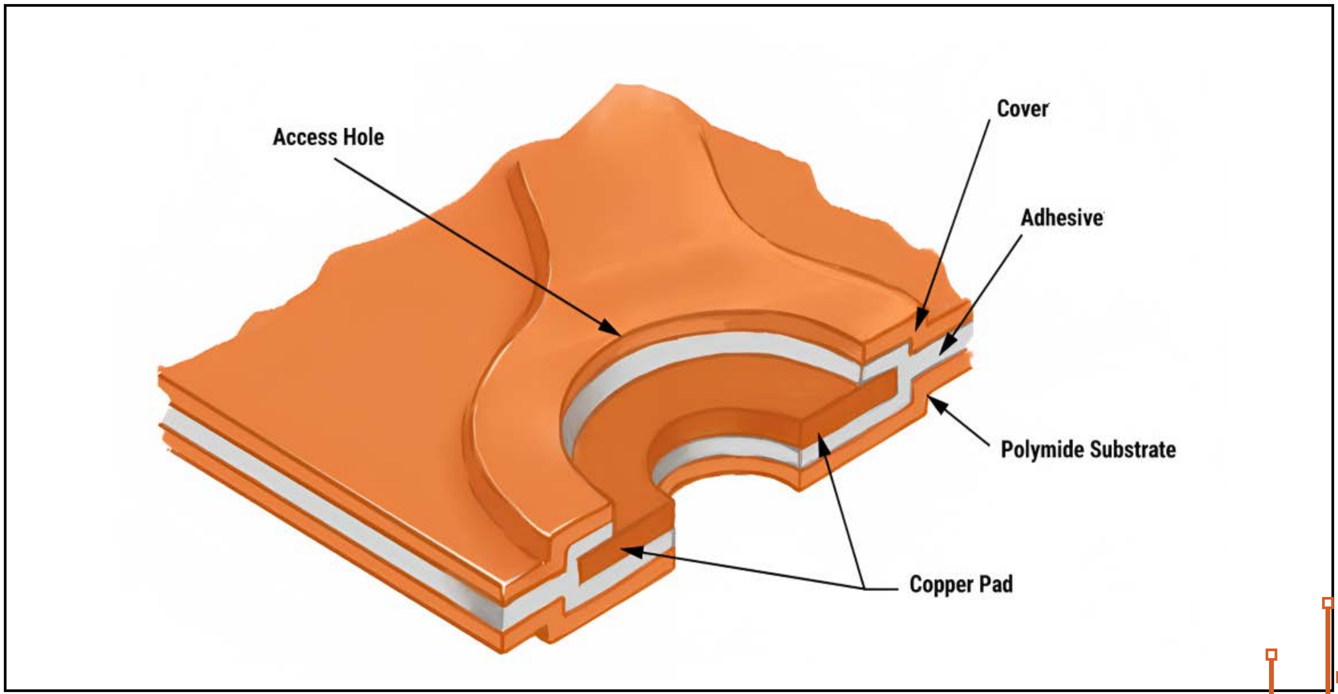


Image 1: Single-sided flex circuit board

Single-sided flex boards are the most basic type. These consist of a single conducting layer on a flex substrate. A flexible polyimide is laminated to a thin sheet of copper. Holes may be drilled through the substrate to allow the component leads to pass through during the soldering process. A polyimide coverlay can be used for circuit insulation and environmental protection.

1.2.2 Double-sided flexible circuit boards

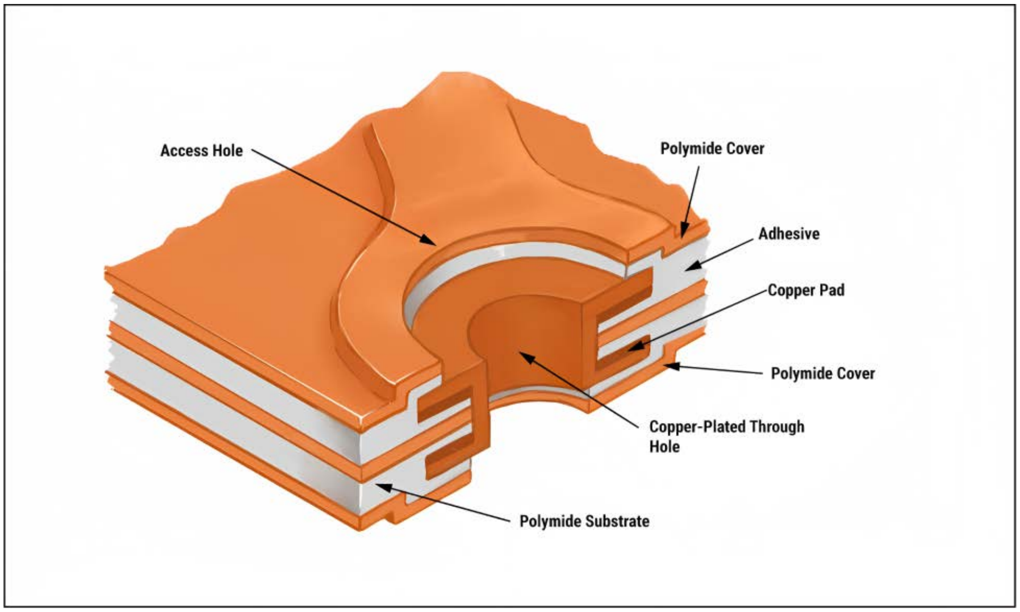


Image 2: Double-sided flex circuit board

Double-sided flex PCBs have two conductive layers (one on each side of the flex substrate). Plated through holes or vias establish electrical connections between the layers.

1.2.3 Multilayer flexible circuit boards

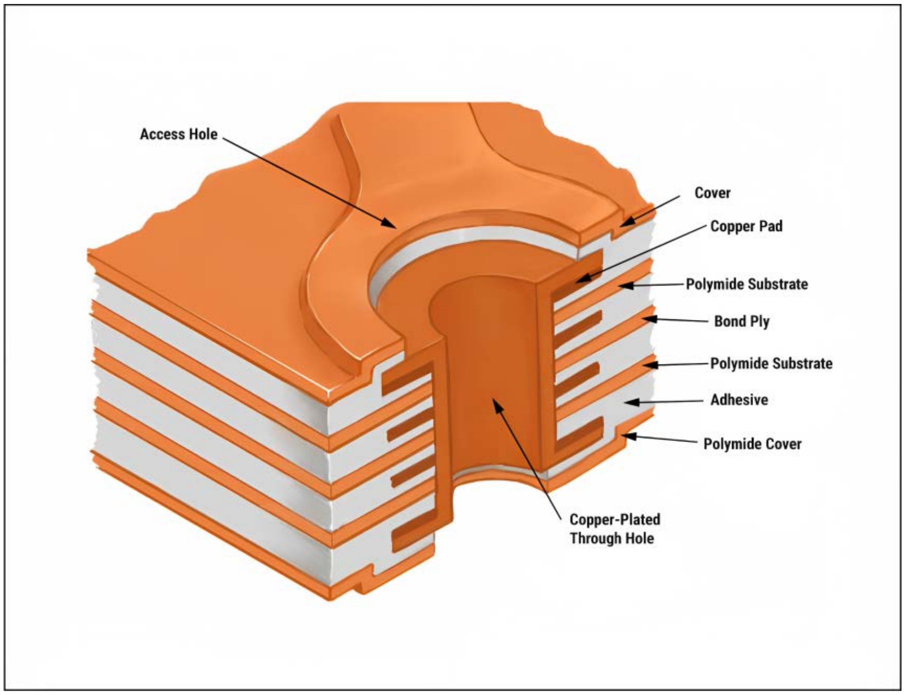


Image 3: Multilayer flex circuit board

Multilayer flex boards consist of more than two copper conductors. Like double-sided flex PCBs, the conductive layers are interconnected through PTHs or vias. These are effective solutions when confronted with design challenges such as unavoidable crossovers, specific impedance requirements, crosstalk issues, additional shielding, and high component density.

1.3 Advantages of flex PCBs

These boards have several advantages over rigid PCBs.

- Flex board can be fabricated in different shapes. This eases the electronic device assembly process.
- Minimizes the connection points. Therefore, it eliminates the chances of interconnection defects like poor solder joints. This makes them more reliable when compared to rigid boards.
- Flexible circuitry is thinner and lightweight than its rigid counterparts.
- Offers superior resistance to vibrations and other disruptions in harsh environments.
- Flex PCB makes use of HDI technology.
- Enables better airflow and heat dissipation than other PCBs.

1.4 Flex PCB materials

Polyimide is the material used for both the flex core and coverlay layers. Flex substrates offer better material properties when compared to standard FR4 rigid materials. The thickness of flex materials is uniform throughout the substrate. These materials also offer improved DK values ranging between **3.2 and 3.4.** The lack of woven glass reinforcement reduces variations in Dk. Typically, the thickness of the flex layers ranges between **1and 5 mil.** Generally, polyimide materials such as Nelco and Rogers are preferred for rigid-flex boards. For flex circuits, Kapton is ideal.

Polyimide flex cores are cladded with rolled annealed copper. This copper is very thin and suitable for both dynamic and static applications. 0.5oz (0.7 mil) or less copper is more commonly used in these boards.

There are two major types of flex materials:

1. **Adhesive-based:** Copper is bonded to the polyimide with acrylic adhesive
2. **Adhesive-less:** Copper is cast directly onto the polyimide

Adhesives are used to laminate the copper layer with the polyimide. The use of adhesives may cause cracks in the copper plating within via holes because acrylic adhesives can become soft when heated. Consequently, when designing for adhesive-based materials, it's important to incorporate anchors and teardrops in your design.

Here are a few disadvantages of using adhesive based materials:

1. Can cause cracks in via holes
2. Makes the copper-clad laminate thicker
3. Prone to absorb moisture from the environment. Hence, it is not suitable for a system that is exposed to the outside environment
4. Leads to dimensional errors as the core thickness may decrease after fabrication

To address these issues, adhesive-less materials are used. Following are the features of adhesive-less materials:

1. Reduced flex thickness
2. Improved flexibility
3. Better controlled impedance
4. Improved temperature rating
5. Well-suited for harsh environmental applications

To choose the right flex material, try our **PCB Material Selector.**

1.5 Applications of flex PCBs

Flexible and rigid-flex PCBs were originally used within the military industry as they require durable, reliable, lightweight 3D circuitry. Now, flex boards are found in nearly every industry. They are used in devices we use on a daily basis—from phones to computers. They are also found in cars, trains, and airplanes to satellites, missiles and radios. In fact, NASA's Mars Rover which is 140 million miles away from the earth has flexible circuits within it.

Chapter 2: How to prepare the rigid-flex board outline

2.1 Board outline

The shape of a PCB depends on the design of the device that it goes into. Once the shape is finalized, test your ideas by cutting out a piece of paper in the shape of your proposed board. Use cardboard to represent stiffeners and rigid areas.

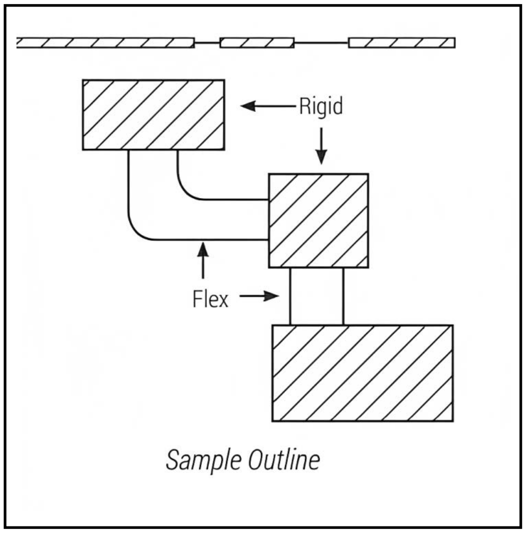


Image 4: Sample board outline

Start your layout by drawing the board outline on a piece of paper. Mark the location of the varied thicknesses. Now, think about preliminary component placement and determine whether those components require stiffeners. Next, mark the stiffeners and rigid areas. This will give you a rough outline of your upcoming board design. It is essential to calculate the bend radius so that thicknesses can be marked precisely. If not planned properly, it will affect your board's flexing capabilities.

2.2 Factors affecting the thickness of flex board

Avoid unnecessary circuit thickness, which hinders flexible capabilities. Determine the flex thickness as per your bend radius requirements. If part of the flex circuit needs to be thicker, add a stiffener.

The following factors determine the required thickness of a circuit.

- Material thickness
- Copper layer count
- Base copper weight
- Adhesive thickness
- Dielectric thickness

Chapter 3: Calculating the bend radius

Bend radius is the measurement of the degree up to which the flex area of a circuit board can bend. It must be identified early in the design phase and calculated based on the number of layers the flex stack-up has. The bend radius ensures your design can survive the required number of bends without damaging the copper.

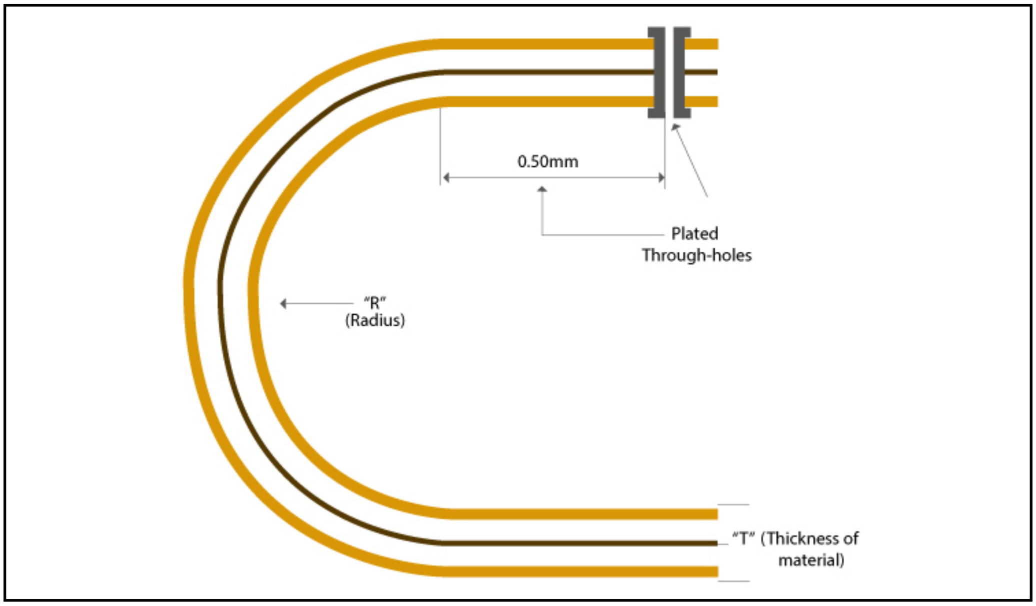


Image 5: Bend radius

3.1 Calculation of bend radius

Number of layers	Bend radius
1 (single-sided)	Flex thickness x 6
2-layer board (double-sided)	Flex thickness x 12
Multilayer board	Flex thickness x 24

***thickness in mils/mm**

It is vital to know two things about flexibility: how many times the PCB will be flexing, and to what extent the PCB will flex. The number of times the flex board can bend determines whether it will be a static or dynamic board.

Note: Plated through holes (PTH) should be at least 0.5 mm away from the bend area, as shown in Image 5.

3.2 Static flex PCBs

A static board is considered bend-to-install and will flex less than 100 times in its lifetime.

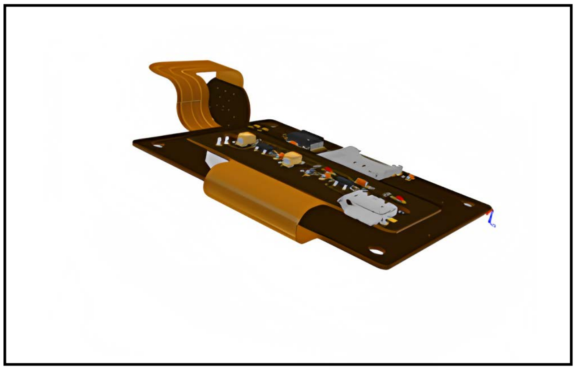


Image 6: Static flex board

It is generally bent during the assembly process. These boards are not intended to flex during the operation of the end product. Static boards may have a slight bend that allows the board to fit into its packaging. The image below shows a board with a static ribbon (flex PCB).

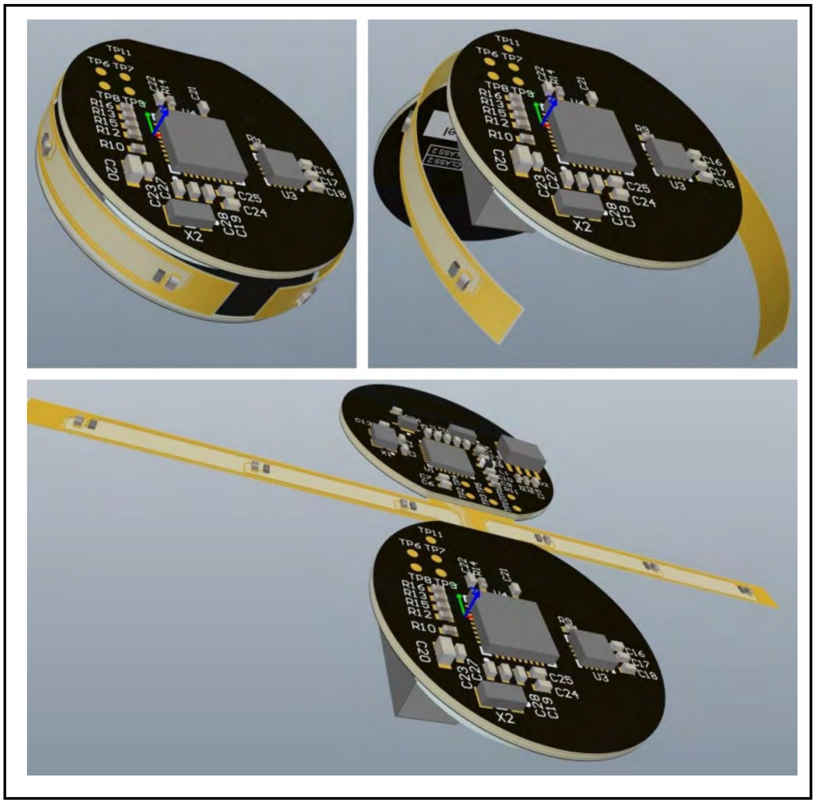


Image 7: PCB with a static ribbon

The flex ribbon in the above example is only bent during the assembly process of the electronic device. Once the board is installed on the end product, the board will remain static.

3.3 Dynamic flex PCBs

A dynamic flex is a board that is regularly flexed during the system operation. This design needs to be more robust and must withstand tens of thousands of bends. These PCBs are used in very harsh conditions, such as in spacecraft and military applications.

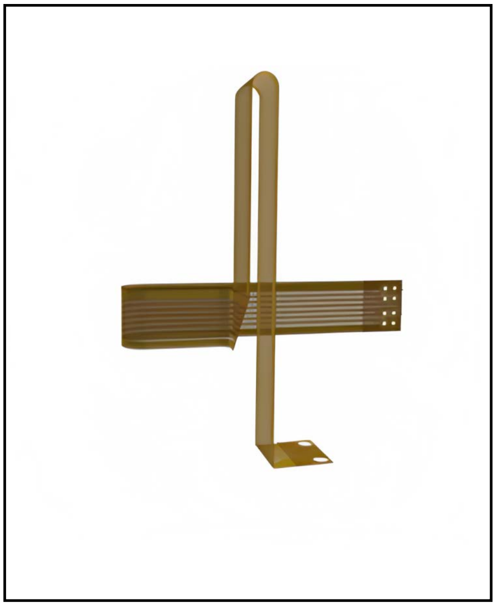


Image 8: Dynamic flex board

3.4 Challenges in dynamic flex PCB design

Dynamic flex boards are capable of solving many interconnect and packaging challenges in applications that require repetitive flexing. The designers should ensure that the copper circuit on these boards does not fracture during operation and result in an open circuit.

3.4.1 Minimum bend radius

The minimum bend radius is one of the critical aspects of design success. The recommended minimum bend radius is **100 times** the finished thickness of the dynamic flex circuit. For example, a flex circuit with a finished thickness of 0.006” will need a 0.6” minimum bend radius or 1.2” minimum bend diameter to ensure its reliability.

3.4.2 Copper type and grain orientation

Flex PCBs generally use rolled annealed copper. Rolled annealed copper (RA copper) is created by subjecting electro-deposited copper to the rolled annealed process. The grain structure is transformed from a vertical to an elongated horizontal structure. This improves the ductility of copper, making it suitable for dynamic applications.

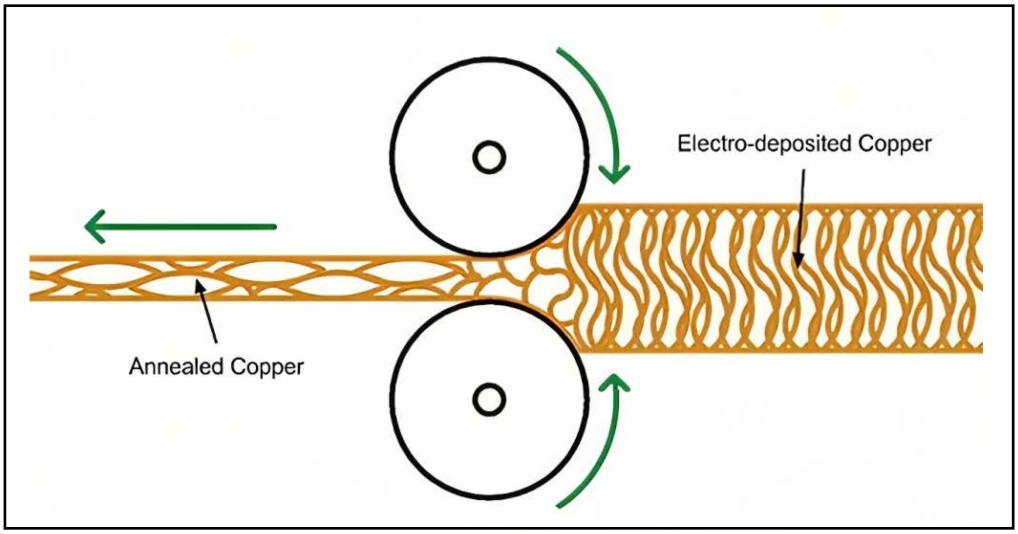


Image 9: Rolled annealed copper

RA copper is available in a variety of thicknesses ranging from ¼ oz up to 2 oz, with ½ oz and 1 oz being the most commonly used.

3.4.3 Layer count and neutral bend axis

Layer count of dynamic flex boards is limited as per the IPC 2223 standard. The optimum configuration is a 1-layer construction. It allows the copper circuits to be close to the neutral bend axis (the neutral bend axis is the plane where there is minimum tension or compression when the circuit is flexed). This ensures that the copper is subjected to the least compressive and tensile forces possible.

A 2-layer construction is permissible provided that a thin adhesive-less flex core of thickness 0.001” or less is used between the two layers. This ensures a minimal distance between the circuits and the neutral bend axis.

3.5 Best practices for designing bend areas

- Avoid 90 degree bends if possible. Tighter bends increase the chance of circuit damage. Gradual bends are safer for the circuit.
- Always measure the bend radius from the inside surface of the bend.
- Place conductors smaller than **10 mil** inside the neutral bend axis, as they tolerate compression better than stretching.

- Avoid plated through-holes within the bend area.
- Keep conductors running through a bend perpendicular to the bend axis
- Use staggered conductors in multilayer circuits to increase the effectiveness of the circuit.
- Provide sufficient space between the transition point of flex and the rigid area to minimize the stress on the flex layers

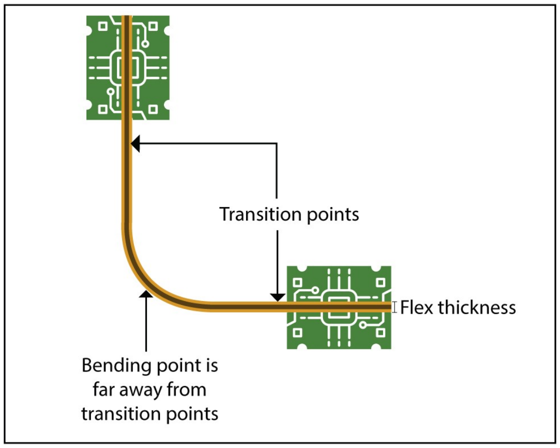


Image 10: Bending point is kept far away from transition points

3.6 Methods for increasing flexibility of flex boards

There are a couple of different methods of increasing the flexibility of a flexible circuit. The most common method is to reduce the overall thickness of the flexible dielectric material because its thickness directly influence bendability.

The second method is to reduce the copper thickness of the traces and moreover the thick-ness of the plane layer. One way of reducing copper on a plane layer is by cross hatching the plane. Typically, we recommend 0.015” wide signals with 0.025” spacing for the cross hatched plane layers.

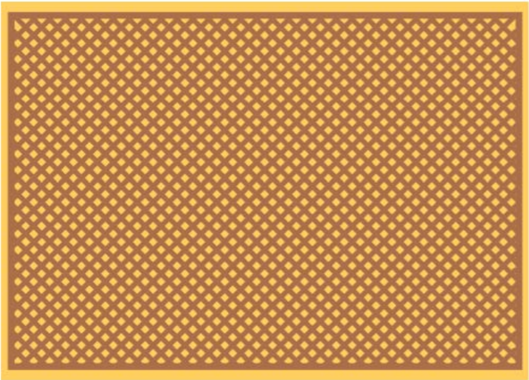


Image 11: Cross-hatched copper plane

Ground and power planes are usually cross hatched in flex PCBs in order to maintain or in-crease the flexibility. A ground or power plane that is completely flooded doesn’t bend.

Chapter 4: Flex routing

Routing is the process of laying copper traces between the nodes. This conductive path is defined by placing tracks, arcs, and vias to establish a connection between two nodes.

The layout of the circuitry has a direct impact on the performance and longevity of a flexible circuit board. Adhere to the following points in mind while routing a flex PCB:

4.1 Guidelines for routing flex PCBs

- Always opt for a larger bend radius.

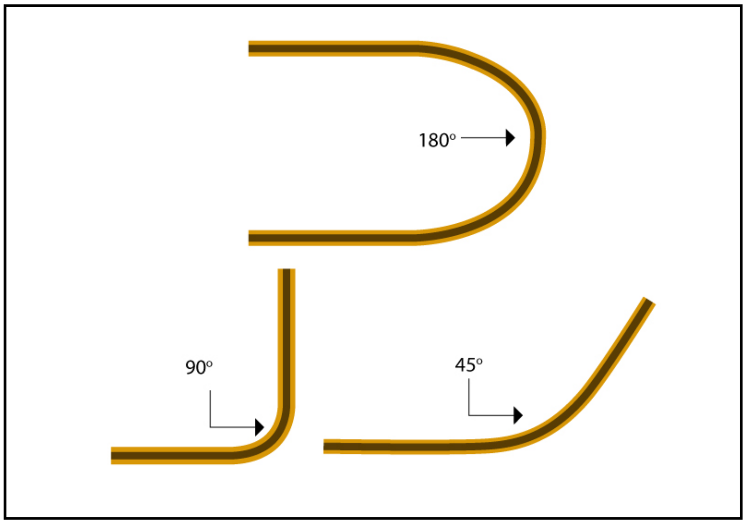


Image 12: Bend radius

- When designing multilayer flexible PCBs, stagger traces on the front and back. Stacked traces will not only reduce the flexibility of your circuit, it will increase stress contributing to the thinning of copper traces at the bend radius.

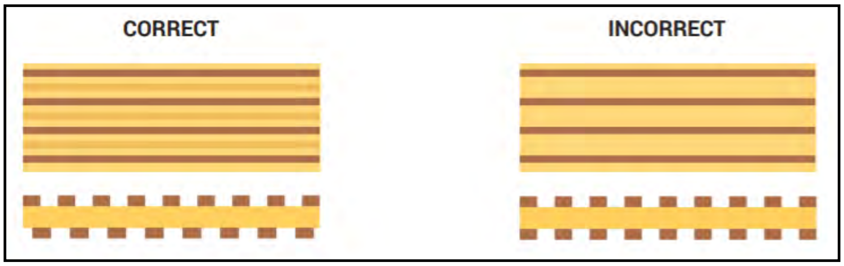


Image 13: Stacked and staggered

- Use curved traces instead of traces with corners.

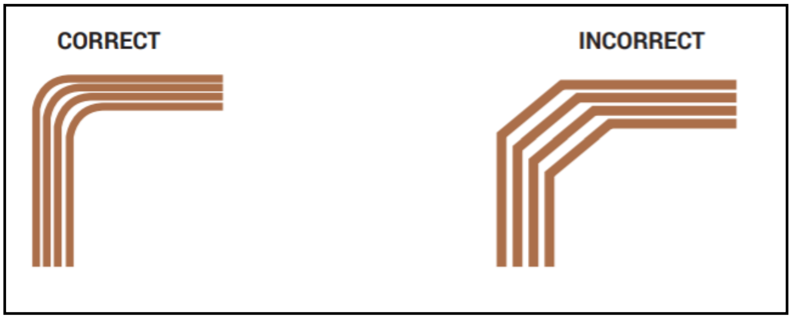


Image 14: Curved and sharp edged

- Traces should be perpendicular to the bend area.

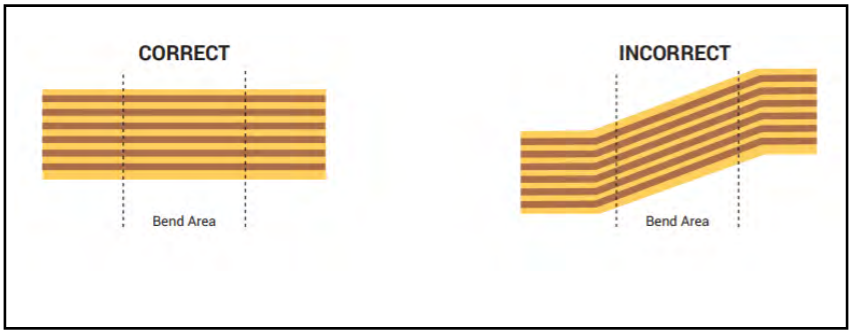


Image 15: Traces perpendicular to bend area

- Trace entering a pad forms a weak spot in which the copper might get fatigued over a period of time. It is always recommended to taper down the pads (as shown below) towards the end at which they are connected to the traces

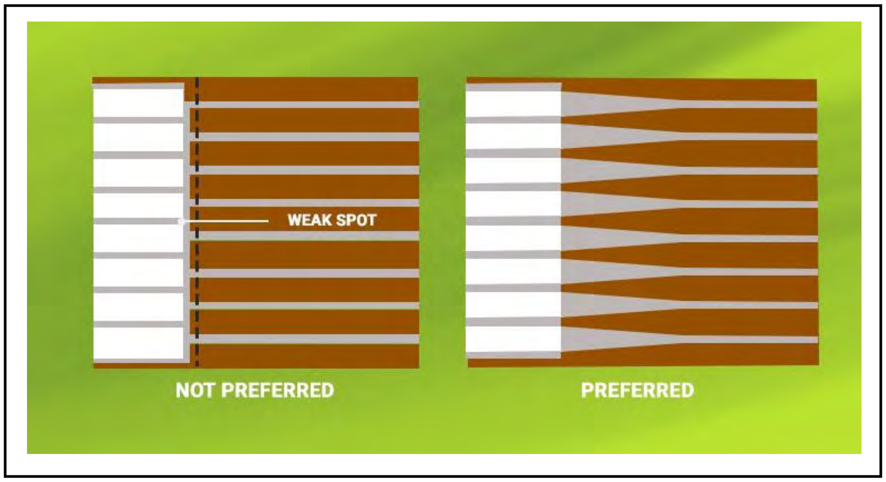


Image 16: Non tapered and tapered pads

- Copper is prone to being detached from the substrate due to the bending of the flex circuit. It is vital to provide mechanical support for the exposed copper. Plating in the through hole vias inherently provides mechanical support in the flex region. For this reason, additional through hole plating of up to 1.5 mil is recommended for rigid-flex and flex circuits. Usually, SMT pads and non-plated through holes are referred to as unsupported and require additional measures to prevent detachment

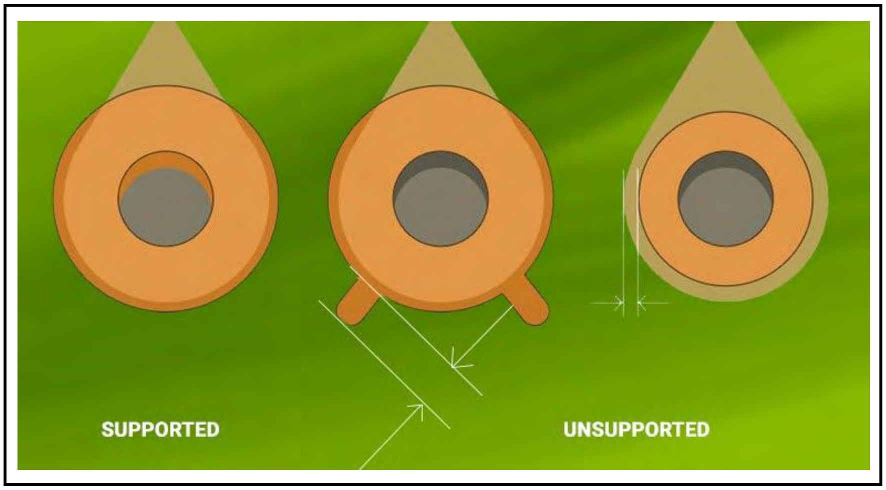


Image 17: Supported and unsupported pads

- Via-in-pad is not recommended in flex designs as that can damage the thin substrate during planarization. Moreover, the smaller aspect ratio of the vias does not allow non-conductive epoxy filling, as it can hamper the electrical conductivity of the vias

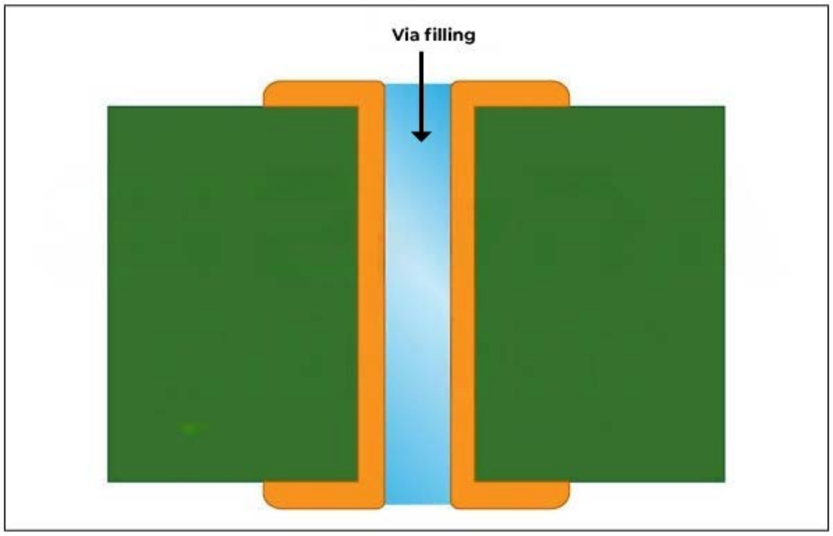


Image 18: Via filling with non-conductive epoxy resin

4.2 Pad design for outer layers

Pads are the exposed regions of metal on a circuit board on which the component leads are soldered. Pads are prone to lifting off due to the flexible nature of the substrate. This can be avoided by incorporating anchors or spurs encapsulated in coverlay. During dynamic bending, anchors help stabilize the outer layer. It is also highly recommended to make the pads as large as possible.

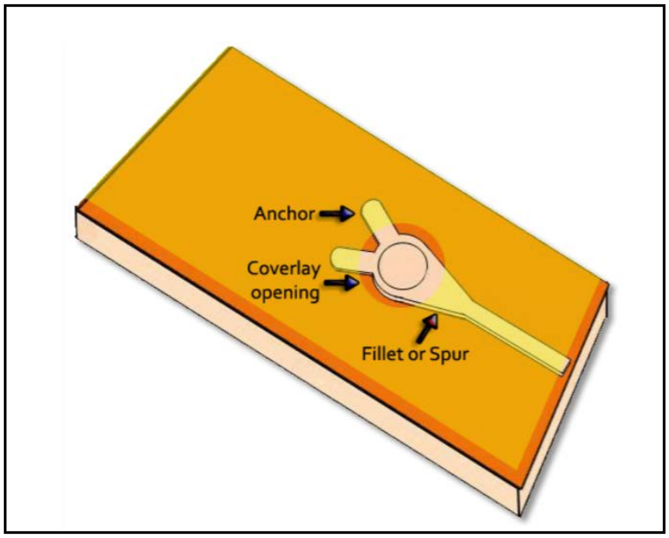


Image 19: Anchored trace on outer flex layer

A teardrop is an extra copper at the junction of a pad and a trace. It is advisable to use teardrops on all flexible PCBs, especially when there is a transition of copper trace from thick to thin. For example, if a part of your trace width is changing from 10 mil to 4 mil, then a teardrop is added at the transition point to reduce any stress or hairline cracks. Teardrops can reduce and even eliminate potential stress concentration points on the PCB.

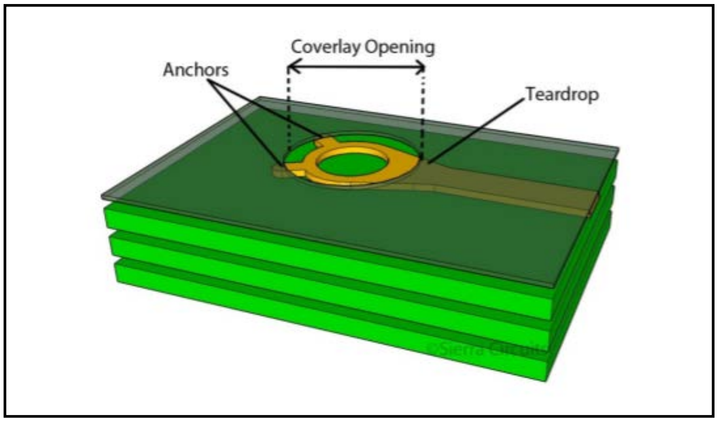


Image 20: Tear dropped trace

Chapter 5: Designing and manufacturing vias in flex PCBs

Most flex PCBs have multiple conductive layers that are electrically connected. The connection through the dielectric layers is established through mechanically drilled or laser drilled vias. IPC 6013 standard defines various considerations related to flex vias. Here are a few points to consider:

5.1 Flex via design

Vias are at greater risk of peeling when implemented in flex designs.To reduce this risk:

- Make the annular rings as large as possible
- Incorporate tear dropping in vias
- Add tabs or anchors to vias to prevent peeling

5.2 Via locations

- Vias are not reliable in areas that will flex and bend
- In a dynamic application, flexed vias can crack very quickly
- Vias are safe over a stiffener, but those that are placed just off its edge are at risk of cracking. The stiffener's edge is kept at least 50 mil away from the vias

5.3 Rigid-flex vias

Hole-to-flex distance is the distance between vias and the rigid-flex transition area. It should be **50 mil** for boards that require high reliability. For commercial applications, the distance can be **30 mil.** Insufficient clearance generates undesired stress during bending and detaches the via from its plating.

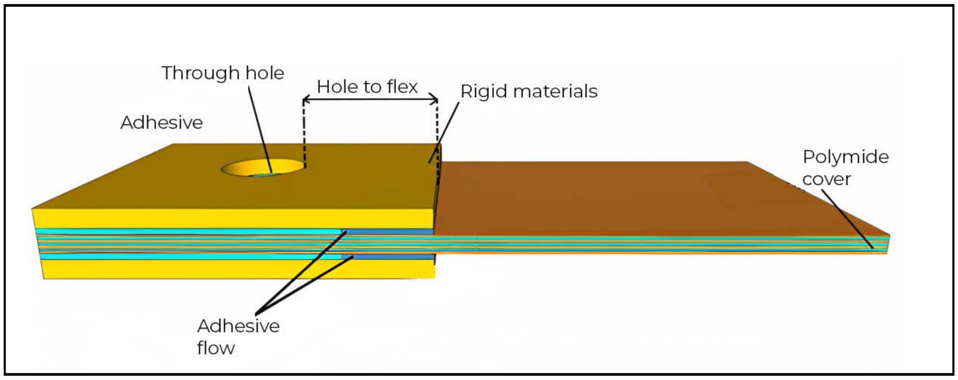


Image 21: Hole to flex distance distance in rigid-flex board

5.4 Annular ring in flex PCBs

Annular ring is the area of copper pad around a drilled and finished hole (copper plated via). There should be enough copper to form a solid connection between the copper traces and the via in a multilayer PCB.

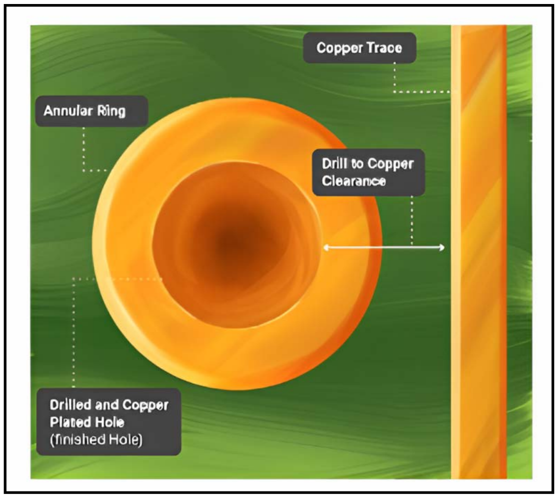


Image 22: Annular ring

The main purpose of an annular ring is to establish a good connection between a via and the copper trace. The minimum annular ring should be **8 mil** for flex PCBs.

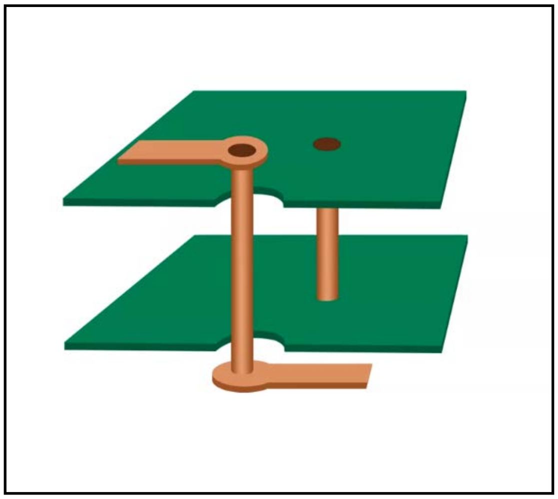


Image 23: Annular ring connection between via and trace

5.5 Plating flex circuit boards

For double-sided flexible boards, Sierra Circuits uses pads only plate" process for plating through holes. To accomplish this, we drill the flexible copper-clad dielectric material and then image the pads around the drilled holes at (drill diameter) D +.003" or better.

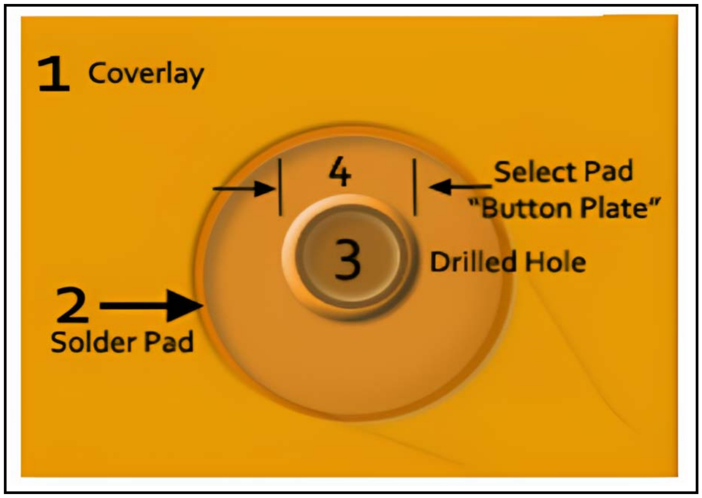


Image 24: Hole plating in flex PCBs

After plating the holes with approximately 0.001" of copper, we again image the final circuitry pattern, and the unwanted copper is etched. An additional D + 0.014" of the pad is needed for this etching process. The table below helps in finalizing the drill size for different layer types.

Layer type	Standard boards (mils)	Advanced boards (mils)
Flex	drill + 14	drill + 10
Outer layer rigid	drill + 10	drill + 6
Inner layer rigid	drill + 14	drill + 10

5.6 Drill to copper

When designing flexible boards, it is crucial to keep drill to copper in mind. Drill to copper is the distance between a hole (via or non plated hole) and the nearest copper feature. Flexible dielectric materials are not as dimensionally stable as standard rigid materials. This material moves, and contracts during the manufacturing process, which makes drill to copper a critical factor while designing flex circuits.

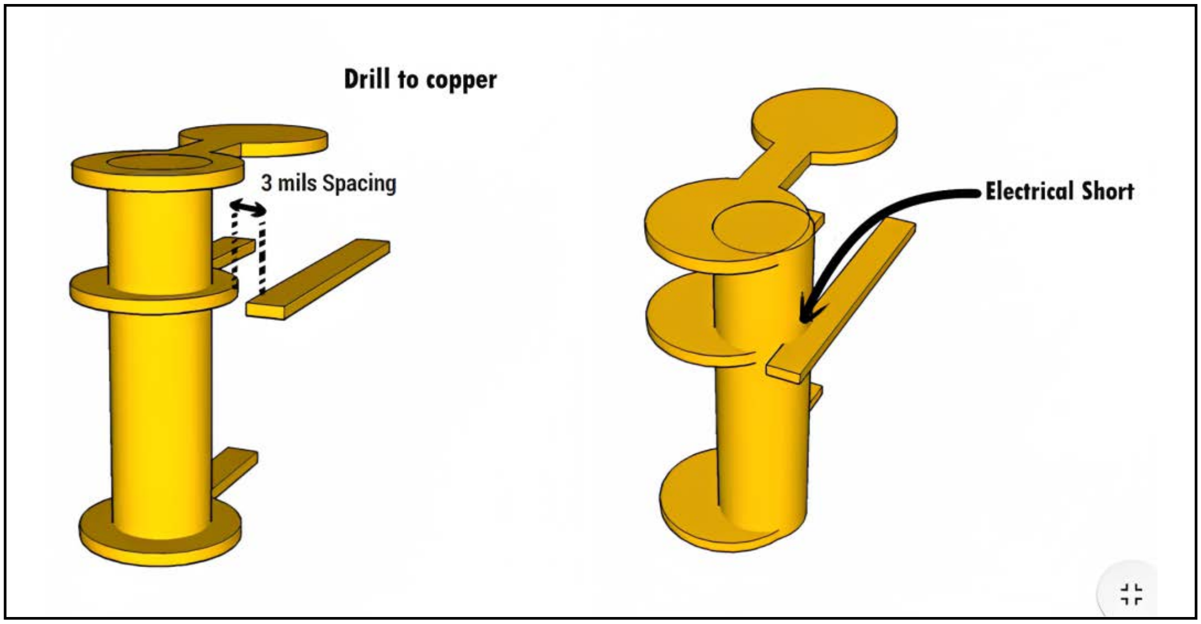


Image 25: Drill to copper

Finished hole is the final hole obtained after metallization and surface finish. The drill to copper clearance and the plating thickness together determine the finished hole-to-copper clearance.

Finished hole to copper clearance = drill to copper clearance + plating thickness x 2.

So, if the drill diameter is 6 mil and the plating thickness is 1 mil, then the finished hole to copper clearance = 6 + 1 x 2 = 8 mils.

For drill to copper, always consider the drilled hole edge. To achieve accuracy in layer alignment, it is important to keep the drill to copper distance around **8 mil.**

To quickly run a DFM check on your designs check out our **Better DFM tool**.

Chapter 6: Designing and assembling rigid-flex PCBs

A rigid-flex PCB is a combination of rigid and flexible materials. Here, one or more flex circuits are used to connect rigid sub-circuits. Rigid-flex designs may be expensive to fabricate, but they can essentially save costs during electronic system assembly.

6.1 How do rigid-flex PCBs cut assembly costs?

Using rigid-flex boards can make way for both direct and indirect cost savings. Direct cost savings mainly comes from reduced **BOM (bill of materials)** and inventory. Indirect cost savings are from reduced assembly costs and improved reliability.
You can quickly validate your BOM files with our **BOM Checker** tool.

Let us assume that a product has 6 interconnected rigid PCBs (a power board, two control boards, and three display boards). Interconnections among these boards would require wire harnesses and connector pairs. Now, let us have a look at how using rigid-flex PCB could reduce direct and indirect costs.

6.1.1 Direct cost savings

A single rigid-flex PCB with 6 rigid sections could be used to replace 6 rigid boards within the electronic device. It also replaces wire harnesses and eliminates connector requirements. This inventory reduction leads to direct cost savings.

6.1.2 Indirect cost savings

Since there are no wiring harnesses involved in rigid-flex PCBs, the cost incurred in procurement and assembly is saved. Also, there are no wiring errors, which increases the reliability of the product. This eases the testing procedure and reduces production costs.

6.2 Materials used in rigid-flex PCBs

Rigid-flex PCBs use a combination of standard rigid and flex materials. The materials include core, prepreg, copper foil, flexible laminates, cover layers, and bond plies. The PCB material used in flex sections can be just a few microns thick but should be reliably etched. This often makes them preferable over rigid PCBs in satellite and aerospace applications.

No-flow prepregs are one of the most critical components in rigid-flex manufacturing. This type of prepreg prevents the flow of epoxy resin onto the flexible sections of your PCB. Flex materials are less dimensionally stable than the rigid materials they are stacked with.

6.3 Rigid-flex design rules

- Place the flex layers in the middle of the stack-up and use an even number of layers
- Drill to copper should be at least 8 mil
- Clearly name the rigid and flex sections
- To allow multiple flex layers to bend without deformation, a bookbinding technique is used where the layers are manufactured in progressively longer lengths around the outside bend radius. The number of flex layers in your design determines the cost of bookbinding. The higher the number of flex layers, the higher the lamination cycles and overall cost. Typically, a board with bookbinding costs 30% more than a non-bookbinding one

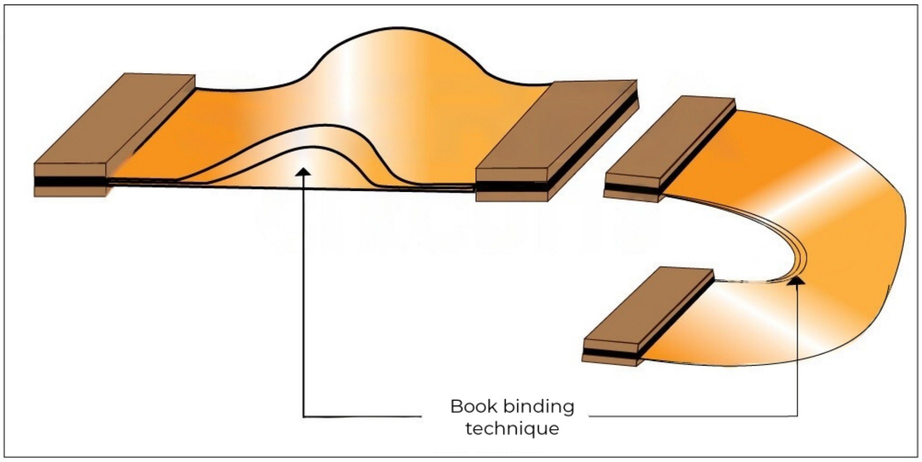


Image 26: Bookbinding construction

6.4 Assembling flex and rigid-flex PCBs

- Solder joints can weaken if components are placed in the bending areas. To solder IC packages on the flex board, incorporate solder bumps. Affixed to the pads, solder bumps offer better control over the connections. You need to ensure the component placement is not infringing on the board thickness restrictions
- If components need to be close to a flex area, add stiffeners. Try to place components over the stiffener or rigid area
- Include SMT components on only one side of the board for flex circuits
- Most parts of the rigid-flex circuits are stiffened, only the hinges or flexible arms remain unstiffened
- Depending on component size, the surface mount areas do not always require a stiffener
- Apply stiffeners to the opposite side of SMT components and to the same side of the connector and through-hole components
- The pre-bake cycle eliminates any retained moisture in the board and allows for improved assembly yields and reliability. However, if boards are assembled immediately after manufacturing, there is no need for pre-baking. Since Sierra Circuits has in-house manufacturing and assembly capabilities, we skip the pre-bake step

6.5 Stiffeners

Single-sided, double-sided, and multilayer flex circuits can be stiffened in specific areas by adding a localized rigid material called stiffener. This can add support for mounting components, increasing strength, thickness, and rigidity. Kapton and FR4 materials are commonly used for stiffeners. These materials can be attached with thermally cured acrylic adhesive or pressure-sensitive adhesive. Stiffeners should overlap the bared coverlay by 0.030to relieve stress. They reinforce solder joints, increase abrasion resistance, and help with strain relief and heat dissipation.

Image 27.1 Stiffener considerations

6.5.1 Stiffener considerations

- Maintain the same stiffener thickness when using multiple stiffeners to lower cost
- Stiffeners should come to at least two edges of the board
- When the desired stiffener thickness is less than 10 mil, use Kapton stiffeners. Generally, their thickness varies between 4 mil and 10 mil.
- The thickness of acrylic adhesives would be roughly 2 mil (50 microns). Although, it may be reduced to 1.6 mil (40 microns) after pressing

6.6 Tear guards

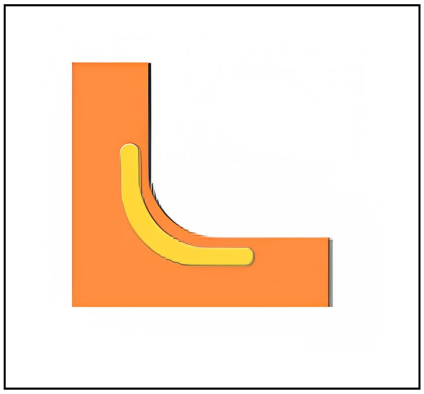


Image 27.2 Tear guards

We recommend the use of tear guards, which will help reinforce the flex material along the inside bend radius. This will help prevent the tearing of the flex material. Avoid any discontinuation of materials close to the bend area and try to use a liberal bend radius, avoiding sharp corners.

6.7 Array panelization and depanelization

Array panelization is a manufacturing process that involves grouping multiple circuit boards into an array. It aids in accurate component assembly and reduces overall costs.

To fulfil the bending requirements of the rigid-flex boards, their components should be rotated, placed on corners, or intertwined. Array panelization allows these configurations on the production panel.

Panelization also eliminates dimensional faults and ensures the correct alignment of the solder paste stencil across the array. As polyimide coverlays involve greater dimensional tolerances, flex and rigid-flex PCB arrays are often smaller than rigid arrays with LPI solder masks. An external frame can be affixed to create enough room to hold the assembly properly. Pressure-sensitive adhesives (PSA) are used to bond flex circuits to the frame.

Always prefer laser depanelization to secure precision, speed, and accuracy. It also reduces mechanical stress on the PCB, resulting in higher throughput than standard depanelization procedures. Avoid V-scoring in flex designs, as the cutting-edge process produces an undesirable rough edge finish.

Chapter 7: Fab drawings for flex PCB

To successfully design a flexible PCB, it is important for the designers to have a basic understanding of the flex drawing requirements. Let us have a look at a few of these.

7.1 Flex PCB stack-up

Flexible PCB stack-up drawing will depict each layer's thickness, including the conductive layers. This should also specify which layers are of rigid material and which layers are of flexible material with copper weights. Sierra Circuits can assist you in designing your flex stack-up.

You can check out our Stackup Designer to get an accurate and detailed PCB stack-up.

7.2 Dimensional drawing and tolerances

The dimensional drawing of a flexible PCB design provides the following information:

- Location of PCB stiffeners.
- Thicknesses of each section of the board and the materials to be used.
- Type of flex board (static or dynamic)
- Locations where the board rarely flexes and frequently flexes.

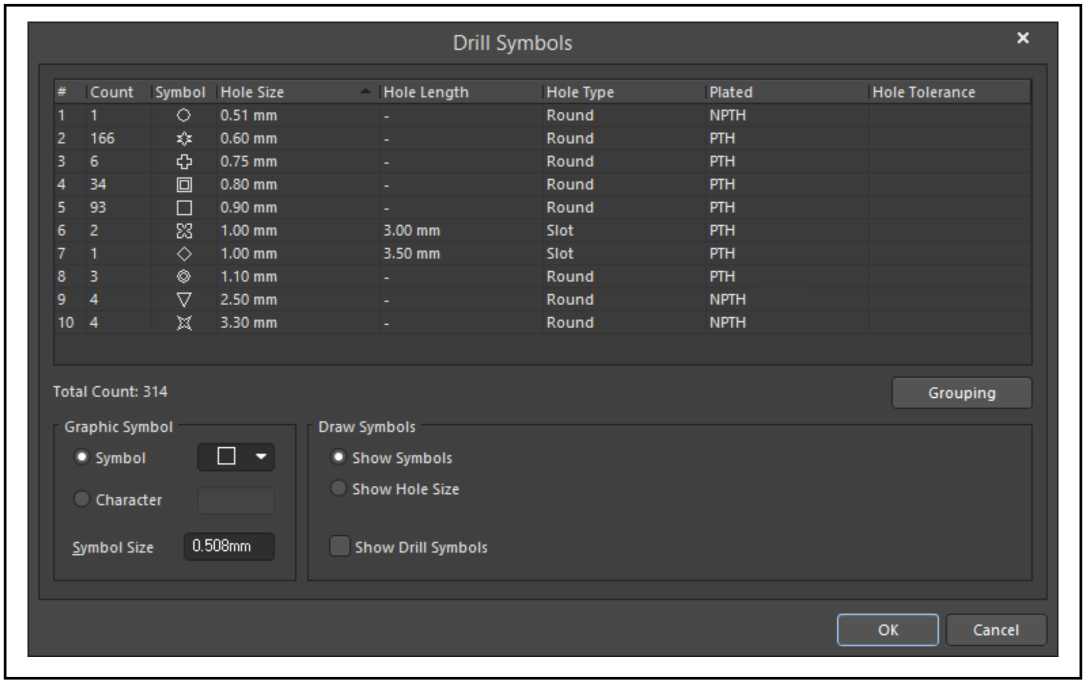
7.3 Design specifications a PCB manufacturer needs to know

A PCB manufacturer expects the following range of specific details from a designer:

- Class type (class 1, class 2, class 3), wiring type, and installation use requirements
- Flexible copper clad material to be used
- The cover coat material
- Minimum conductor width and spacing
- Maximum board thickness
- The minimum size of plated through holes
- Electrical test requirements
- Color of coverlay
- Color of silkscreen
- Board markings such as part number, version, and company logo
- Special packaging and shipping requirements

7.4 Drill symbol chart

The drill symbol chart summarizes the drill hole information of the board. An example of a drill symbol chart is shown below. The standard finished hole size is +/- 0.003" but this measurement must be specified on your design drawing.



#	Count	Symbol	Hole Size	Hole Length	Hole Type	Plated	Hole Tolerance
1	1		0.51 mm	-	Round	NPTH	
2	166		0.60 mm	-	Round	PTH	
3	6		0.75 mm	-	Round	PTH	
4	34		0.80 mm	-	Round	PTH	
5	93		0.90 mm	-	Round	PTH	
6	2		1.00 mm	3.00 mm	Slot	PTH	
7	1		1.00 mm	3.50 mm	Slot	PTH	
8	3		1.10 mm	-	Round	PTH	
9	4		2.50 mm	-	Round	NPTH	
10	4		3.30 mm	-	Round	NPTH	

Image 28: Drill symbol chart

7.5 Flexibility (bend radius)

The flexibility of a flex PCB is determined by the bend radius of the flex material used. Bend radius is the minimum angle at which the flex region can bend. The number of bends your flex PCB may experience is an important design consideration. The copper will stretch and crack if the board bends more than its limit.

7.6 Plating requirements

Plating the through holes, or vias, is a requirement for multilayer PCBs. It is very important to mention the type of plating that your flex board requires. The available types of plating are:

- **Panel plating:** This method of plating deposits copper on the entire panel. Panel plating is generally performed before circuit imaging.
- **Pattern plating:** This type of plating deposits copper on the selected areas of flexPCB.
- **Pads only plating:** It is a type of pattern plating where a photoresist covers the entire panel, exposing the pads around the vias. As a result, only the vias and exposed pads get plated.

7.7 Testing requirements

PCBs undergo various tests before they reach the designer. It is important to mention the physical and electrical testing requirements (test type and frequency). Over-specifying the test requirements might increase the overall circuit cost. Basic tests include:

- Dimensional checks
- Electrical continuity
- Ionic cleanliness testing
- Flexibility check
- Plating thickness
- Insulation resistance

7.8 Marking requirements

Designers can specify the type of ink to be used for various board markings like serial numbering, component mounting locations, stiffeners/cover locations, and panel-based marking. The types of ink include:

- Durable white ink
- Traditional epoxy ink

7.9 What to include in flex drawing notes

- The PCB shall be fabricated to IPC-6013, class <1/2/3>, wiring type <enter your requirement here>, and installation use <enter your requirement here>
- The flexible copper clad material shall be IPC-FC-241/11 preferred (or /1) (flexible adhesive-less copper clad dielectric material). For example, FR / AP / LF
- The covercoat material shall be as per IPC 4203/1 <your requirement here>. Example: LPI or coverlay
- The maximum board thickness shall not exceed <your requirement here> and applies after all lamination and plating processes
- For rigid-flex constructions, the acrylic adhesive thickness through the rigid portion of the panel shall not exceed 10% of the overall construction
- Minimum thickness of plated through-holes is 0.001”, with a minimum annular ring of 2 mil
- Misregistration between any two layers shall not exceed ±0.005”
- Warpage shall not exceed 0.010 inch per inch
- The plating thickness of PTHs shall be at least 0.001 mm
- If polyimide material is used in a rigid section, the rigid material shall be GIN (Poly) as per IPC4101/40
- If the thickness requirement is critical, then mention it in the drawing notes
- All external conductive surfaces not covered by a solder mask shall be plated with ENIG
- Provide marking and identification requirements, packaging, and shipping data
- Specify electrical test procedures

- Apply a green LPI solder mask to the rigid sections of the board over bare copper on both sides
- All exposed metal will have <surface finish requirement here. For flex circuits, use a 1 mil polyimide coverlay
- For flex boards, choose ENIG as the surface finish. You can also use immersion gold with 125 microns of nickel and 2 microns of gold. Due to the low lead solderability, a tin or lead finish is not advised for flex material

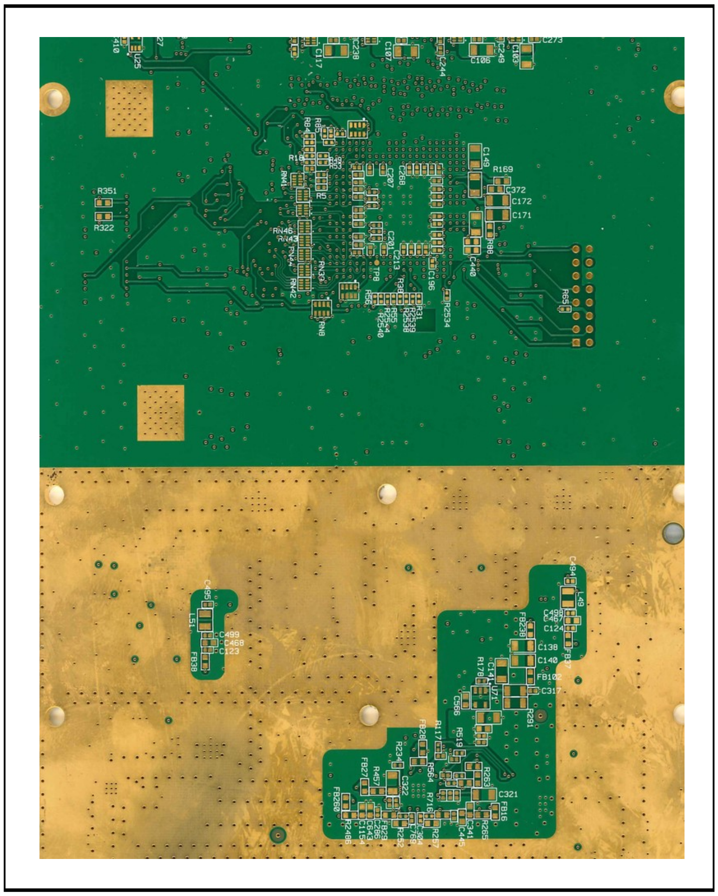


Image 30. ENIG PCB surface finish

- Add silkscreen markings on both sides of the board using white, non-conductive epoxy ink
- The PCB shall be constructed to meet a minimum flammability rating of V-0

7.10 Flex PCB design checklist

In order to get the most out of your flex PCB, you should have a clear vision of the circuit board's functionality and design rules. Below are some guidelines that we have discussed throughout the course of this design guide.

- Make annular rings as large as possible
- Vias should be tear-dropped. Teardrops can reduce potential stress concentration points on the PCB
- Adding tabs or anchors to vias will help prevent peeling
- Vias are not reliable in areas that will bend
- Avoid vias in the flex section of dynamic boards as they are at risk of cracking
- Vias can be used over a stiffener, but they are at risk of cracking if placed at the edge of a stiffener
- Vias should be placed at least 30 mil away from the rigid-flex/flex interface
- Always opt for a larger bend radius
- Use curved traces instead of traces with corners. Curved traces cause lower stress than angled ones
- Maintain at least 10 mil clearance between two flex regions

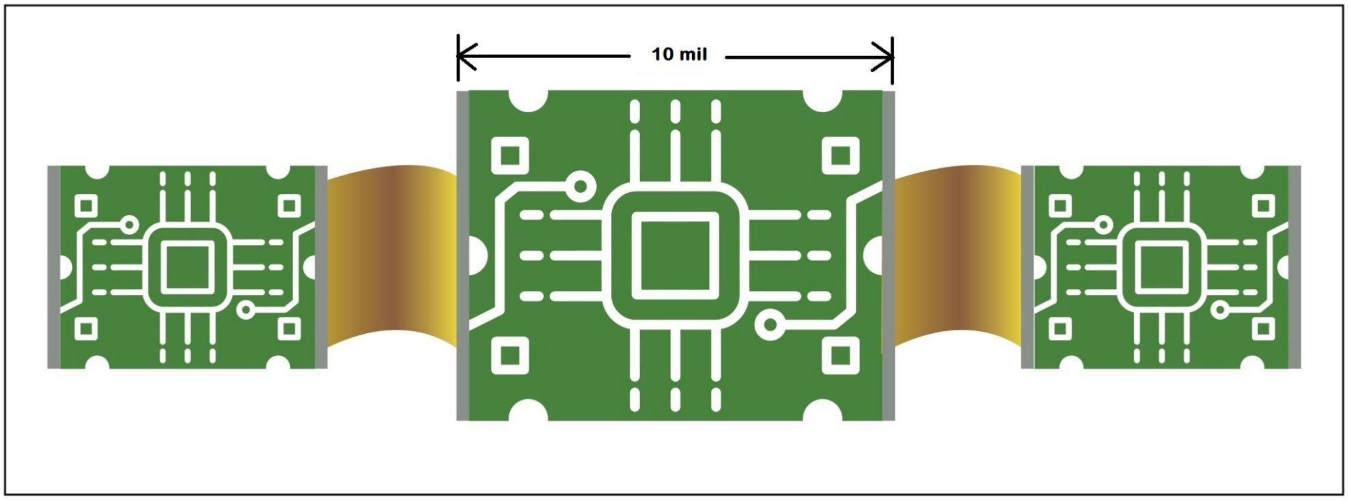


Image 31. Distance between two flex regions of a rigid-flex board

- When designing multilayer flexible PCBs, incorporate staggered traces on the front and back. Stacked traces will not only reduce the flexibility of the circuit but also increase stress, contributing to the thinning of copper circuits at the bend radius
- Traces should also be kept perpendicular to the overall bend
- The rigid-flex fab notes must consist of rigid notes, and flex notes separately
- The acrylic adhesive thickness through the rigid portion of the panel shall not exceed 10% of the overall construction
- Misregistration between any two layers shall not exceed ±0.005''
- Warpage shall not surpass 0.75%
- Provide impedance trace details such as trace width, height, and impedance tolerance in the stack-up

Chapter 8: Controlled impedance in flex PCBs

8.1 What is controlled impedance?

Controlled Impedance is the characteristic impedance of a transmission line formed by PCB traces and its associated reference plane. It becomes very important when high-speed signals propagate on circuit board transmission lines. In order to achieve good signal integrity, a uniform controlled impedance is required.

Type of transmission line	Impedance value	Line width	Line spacing
Single-ended	50 Ω	4 mil	-----
Differential pair	90 Ω	4 mil	5.5 mil
Differential pair	100 Ω	4 mil	6 mil
Differential pair	120 Ω	3.7 mil	7.5 mil

8.2 Factors affecting impedance control in flex

Controlled impedance is determined by the physical dimensions of the PCB traces and the properties of the dielectric material used. Below are the factors that affect impedance control on flex boards:

8.2.1 Physical dimensions of the traces

1. Trace height
2. Trace width at the top and bottom
3. Difference between the width at the top of the trace and the bottom of the trace

8.2.2 Dielectric properties of the material use

1. Dielectric constant
2. Dielectric height between the trace and the reference plane

8.3 Controlled impedance configurations for flex circuit boards

Common controlled impedance configurations for flex circuit boards are:

8.3.1 Single-ended microstrip

It is the most preferred controlled impedance configuration for flex circuit boards. It has a transmission line of a single uniform conductor on the outer layer of the board. The return path for the signal is provided by a reference plane separated by a dielectric lay-er of height H1. This configuration allows for thinner flex construction, improved bend capability and reduced cost.

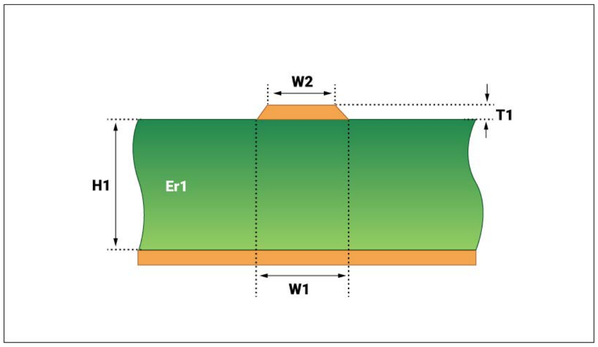


Image 32: Single ended microstrip

8.3.2 Edge coupled coated differential pair microstrip

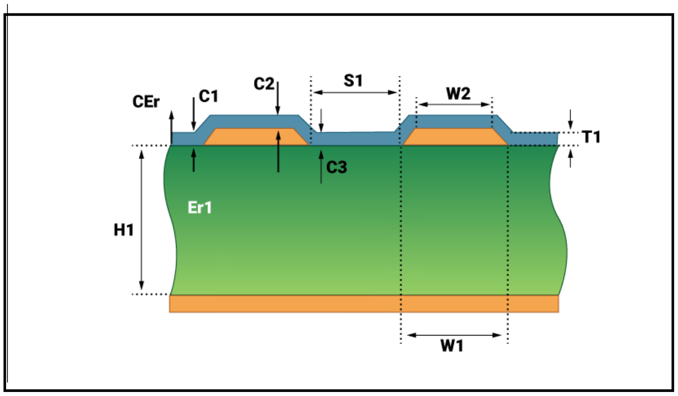


Image 33: Edge coupled coated differential pair microstrip

When a signal and its complement are transmitted on two separate traces, it is called differential signaling. These traces are called differential pairs. The traces are routed with a constant space between them. One of the primary advantages of having edge-coupled differential pairs is that the noise on the reference plane is common to both traces. This cancels out the noise at the receiver end.

This technique is used for routing differential pairs and has the same arrangement as regular microstrip routing. It is more complex due to the additional trace spacing requirements. It consists of a differential configuration with two controlled impedance traces on the surface, separated by a uniform distance and backed by a plane on the other side of the laminate.

8.3.3 Single-ended stripline

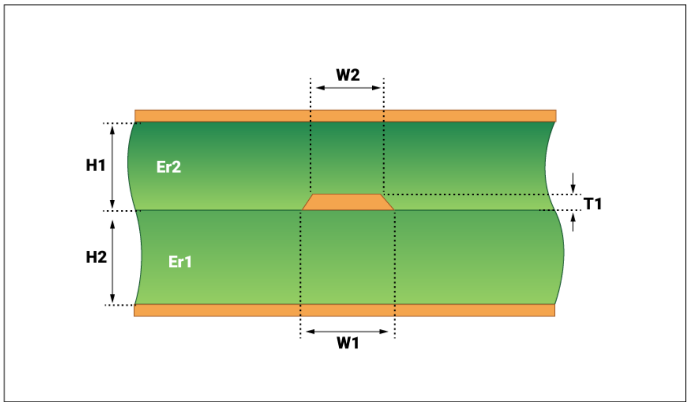


Image 34: Single ended stripline

It implements the signal trace between two ground planes in a multi-layer PCB. The return path for a high-frequency signal trace is located above and below the signal trace on the planes.

8.3.4 Edge coupled differential stripline

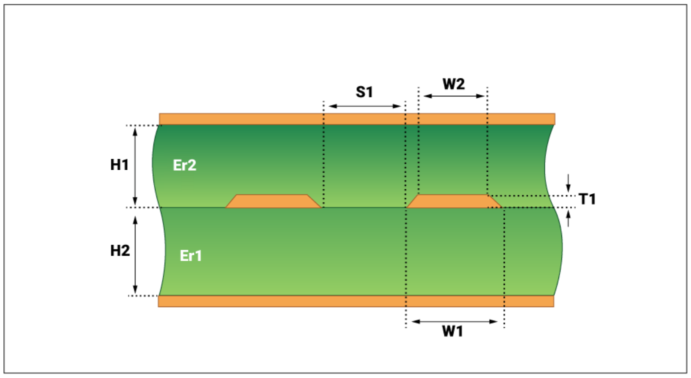


Image 35: Edge coupled differential stripline

It is similar to the single-ended stripline described above, except that we now have a pair of conductors separated by a uniform distance. It is a differential configuration with two controlled impedance traces sandwiched between two planes.

Reference planes in flex PCBs

Cross-hatched copper planes are used as reference planes in flex circuit boards. An image of a cross-hatched plane is shown below.

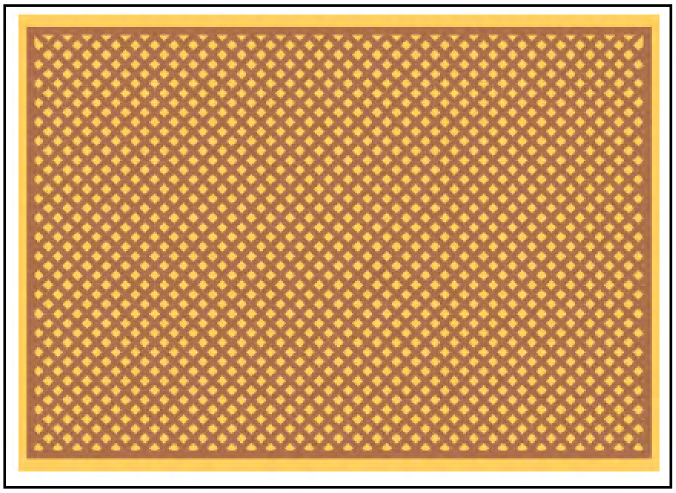


Image 36: Cross-hatched copper plane

A cross-hatch plane can be characterized by the ratio of the cross-hatch conductor width (HW) to the cross-hatch pitch (HP). The lower the ratio, the greater the percentage of copper being removed. 50% copper removal can be achieved if the ratio is about 0.293. The controlled impedance increases proportionately to the amount of copper that is removed in the cross-hatch.

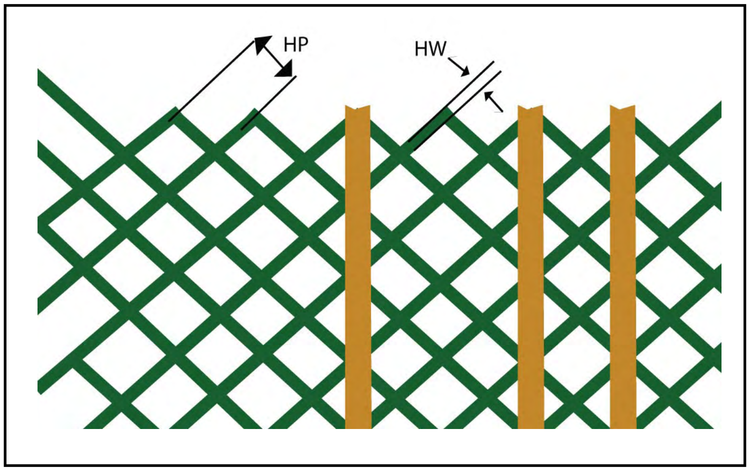


Image 37: Hatch pitch and hatch width in flex reference

8.4 Flex PCB materials for controlled impedance

Flex PCBs are often made of polyimide flex materials. These materials are well suited for controlled impedance designs as they offer a lower Dk value, tightly controlled thickness, and homogeneous material construction.

There are two types of polyimide materials: adhesive-based and adhesive-less materials. Both can be used for the controlled impedance designs. However, adhesive-less materials are preferred for high-speed applications due to their consistent results.

Advanced materials like Teflon and Teflon/Polyimide hybrids are suitable for this. These materials are more expensive than polyimide materials.

It should also be noted that standard adhesive-less polyimide materials meet the controlled impedance design requirements while reducing overall costs.

Sierra Circuits uses **Dupont** for flex PCBs.

Chapter 9: IPC standards for flex PCBs

9.1 Design

IPC-FC-2221	Generic standard on printed board design
IPC-FC-2222	Sectional design standard for rigid organic printed boards
IPC-FC-2223	Sectional design standard for flexible printed boards

9.2 Materials

IPC-4202	Flexible base dielectrics for use in flexible printed circuitry
IPC-4203	Adhesive coated dielectric films for use as cover sheets for flexible printed circuitry and flexible adhesive bonding films
IPC-4204	Flexible metal-clad dielectrics

9.3 Performance

IPC-6011	Generic performance specification for printed boards
IPC-6012	Qualification and performance specification for rigid printed boards
IPC-6013	Qualification and performance specification for flexible printed boards

9.4 Quality guidelines — circuits & assembly

IPC-A-600	Acceptability of circuit boards
IPC-A-610	Acceptability of printed circuit board assemblies
IPC/EIA J-STD001	Requirements for soldered electrical and electronic assemblies

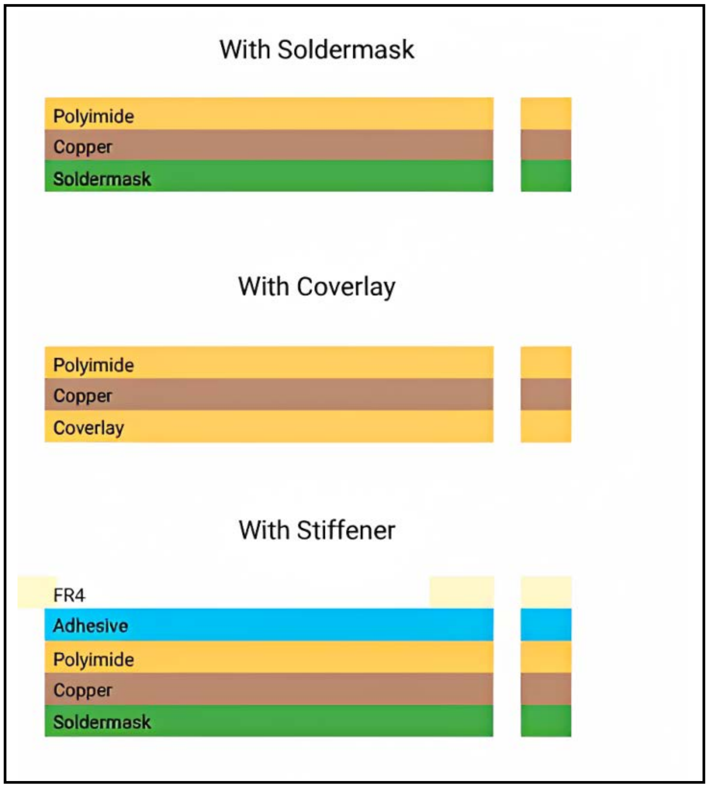
9.5 Military

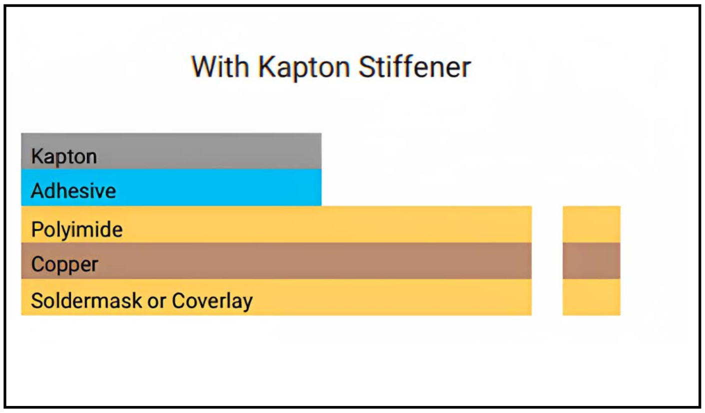
MIL-P-50884	Military specification: Printed wiring board, flexible or rigid-flex
MIL-PRF-31032	Performance specification: Printed circuit board/Printed wiring board, general specification

IPC-6013 Class C meets the same performance requirements as MIL-PRF-31032, and is accepted by government agencies as a COTS equivalent of the latter. If your flex circuit must meet performance requirements of MIL-P-50884, MIL-PRF-31032 or IPC6013, follow the IPC-2223 design specification recommendations.

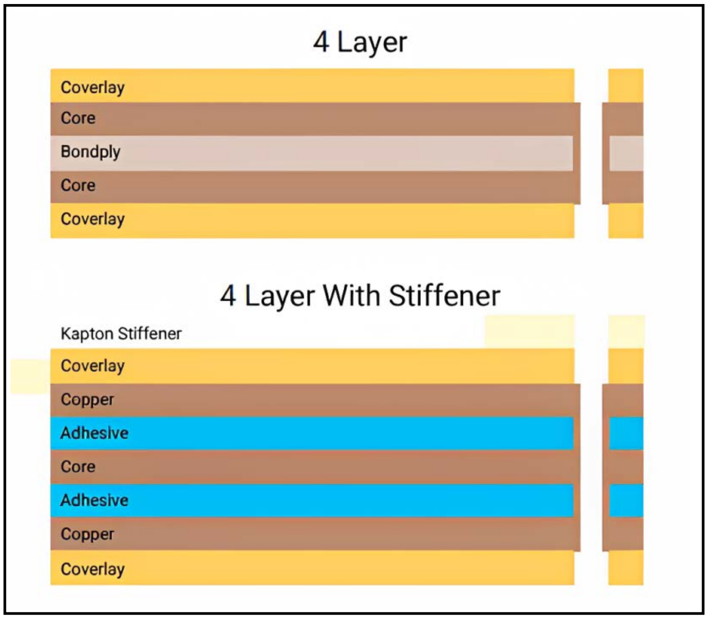
Chapter 10: Example flex stack-ups

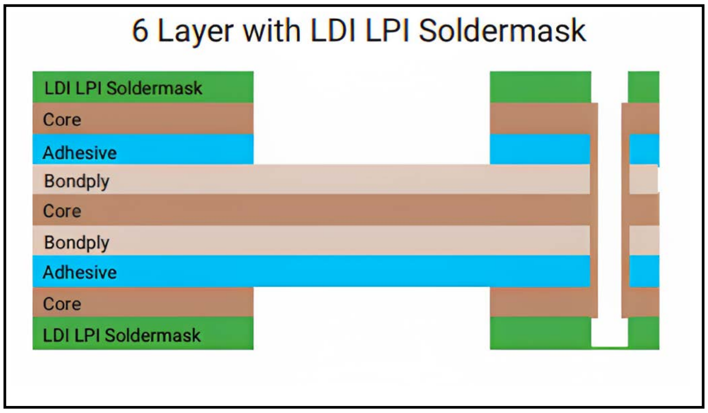
10.1 One layer flex stack-ups



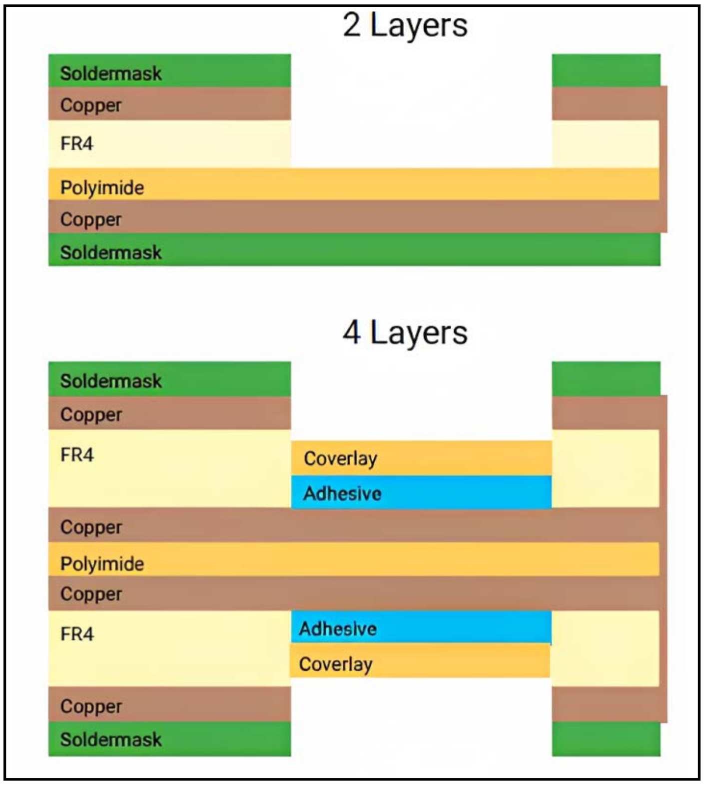


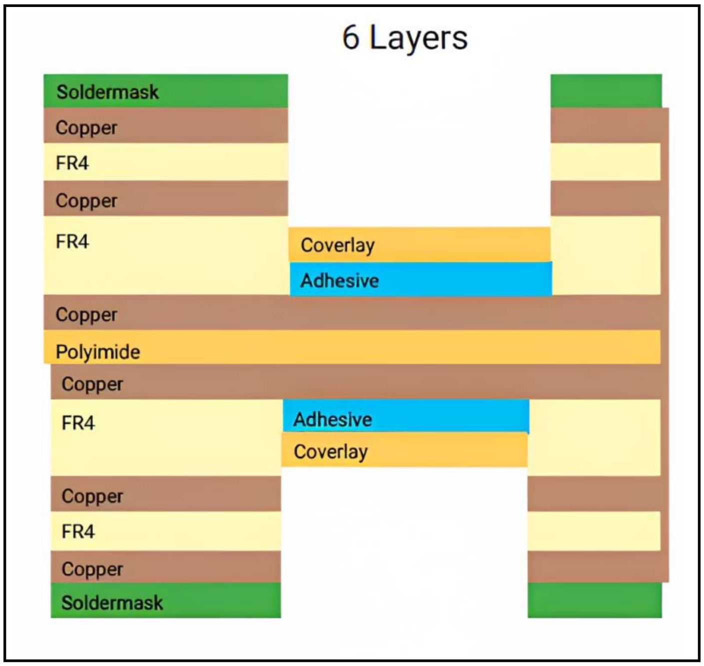
10.2 Multilayer flex stack-ups





10.3 Rigid-flex stack-ups





A well-designed flex PCB will be lightweight, durable, easy to install, and suitable for demanding applications such as wearable devices and satellites. The physical advantages of flex are that it offers improved resistance to shock and vibrations and better performance in harsh environments.

We provide our customers with unprecedented quality, reliability, and a single point of support. No more miscommunication between multiple vendors and no more delays.

We are **ISO 9001:2015, ISO 13485:2016, and MIL-PRF-31032 / 3 Flex** certified.

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